IOWA STATE UNIVERSITY

Knowledge-Grounded 3D Disease Editing: What Does 3D Add Beyond 2D?

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Pilot results and three focused tests of what 3D adds
Advisor update

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The Question We Have to Answer

What does knowledge-guided 3D give us that the original 2D photo, or a knowledge-guided 2D edit, does not?

01 **Not enough: 2D → 3D → 2D**
Rendering a reconstruction back to the input view adds nothing the photo did not have.

02 **Candidate gains**
Shape symptoms (curl, pucker, galls), hidden surfaces (leaf underside), views a camera never took, one consistent object across views.

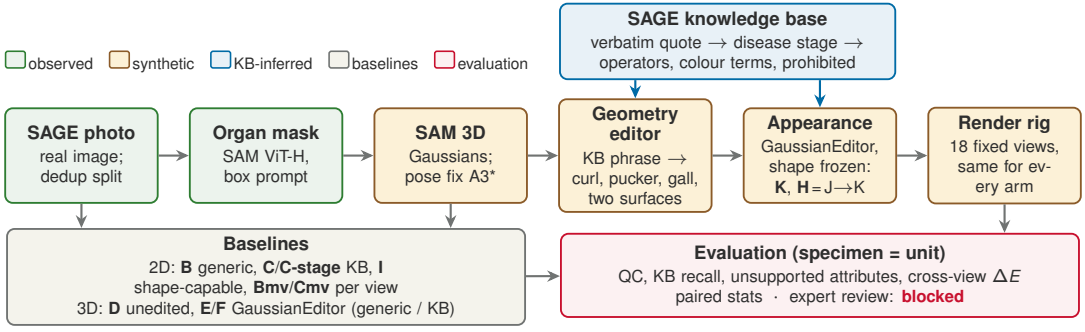
03 **Our rule**
Every claim must be measured, and the measurement must say whether it is *real*, *synthetic* or *unverified*.

How this deck is organised

Act I. Question, five hypotheses, pipeline, status.
Act II. Protocol: audit, case selection, data tiers, structured KB, matched arms.
Act III. Experiments E1–E5, three focused tests of what 3D adds, ablations, statistics, failures.
Act IV. Go / no-go, answers to the 12 questions, next steps.

Status. Pilot runs complete on all 15 specimens; three focused tests added. Numbers carry their *n* and are regenerated from the result files (2026-10-15 04:22).

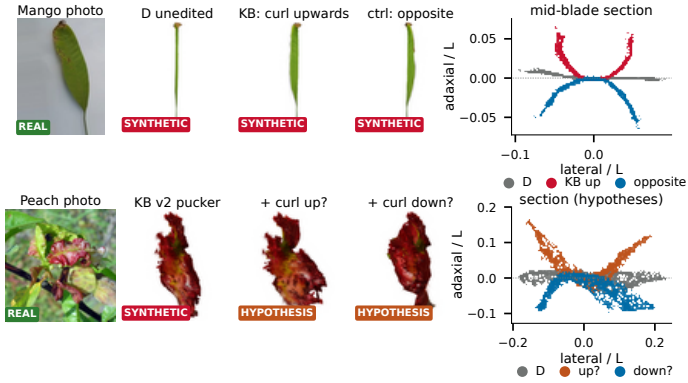
Pipeline as Implemented



Two editors. GaussianEditor changes colour, not shape (E2); shape comes from our *geometry editor* (KB terms without an operator are logged, never approximated). Three streams stay separate: observed (kept from the photo), KB-inferred (a cited phrase), synthetic (filled by a model, unverified).

OUR RUN – MEASURED diagram traced from the code paths (src/figures/fig1_pipeline.py lists them). SAM 3D Objects, GaussianEditor (Chen *et al.*, CVPR 2024), InstructPix2Pix (Brooks *et al.*, CVPR 2023).

Test 1: Directional Shape From the Text



Positive control: the operator follows a direction

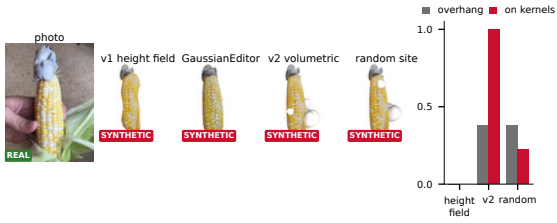
Counterfactual (leaves have anthracnose): "curl upwards" raises margins 3/3 (4.8° →46.2°), reversed 3/3 (-35.4°). IoU 0.91→0.62: contradicts the photo. Verifies code, not anatomy.

Peach: the source gives no direction

The operator **abstains**; up/down only as labelled hypotheses. Pucker on reddish tissue: enrichment 1.14 vs 1.07 uniform.

SYNTHETIC – UNVERIFIED **OUR RUN – MEASURED** results/demos/{mango_curl,peach}/*/metrics.json; roll = signed margin angle about the midrib (+ = toward the upper side). Recovering an imposed angle is a compliance check; a real curl phenotype is still needed.

Test 2: Galls as Volumes on Kernel Tissue



What a height field cannot do

v1 only pushes the surface out (no overhang). v2 galls are closed lobed bodies; 0.38 of their surface overhangs the ear.

Placement from evidence

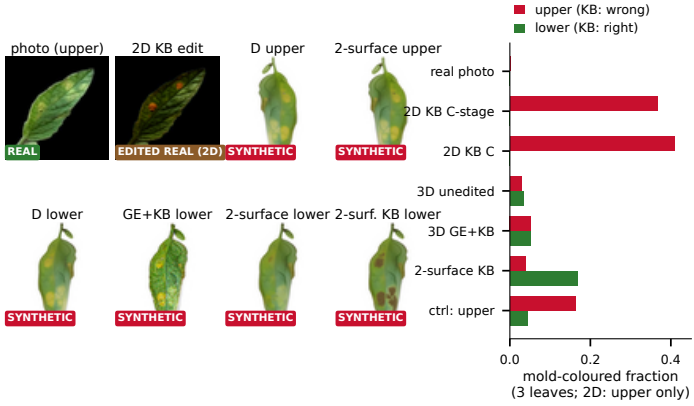
"...develop on the ear replacing kernels" (UNL): on kernel tissue in 1.00 of runs vs 0.22 random: *constraint compliance*, not a finding.

What is assumed

Ear length 18 cm (no ruler); count and position are in no source. 1 ear abstained (no kernel tissue).

REAL SAGE DATA **SYNTHETIC – UNVERIFIED** **OUR RUN – MEASURED** results/demos/corn/*/metrics.json, n = 2 ears × 3 seeds. Added volume 6.1 vs 4.5‰ of (ear length)^3 (galls vs height field; scale-free); 35.7 vs 26.4 cm^3 only under the assumed 18 cm ear. Not validated against real galls.

Test 3: Upper vs Lower Leaf Surface



2D puts the mold on the wrong side

KB: mold “on the lower leaf surface below leaf spots”. A photo shows the upper side only; KB 2D edits paint mold colour on 17–49% (C-stage) / 31–53% (C) of it; real: 0–0%.

Two surfaces keep it apart

One sheet mirrors upper spots below (12–20% of lower pixels). Two surfaces: mold 7–29% of lower pixels, upper 0–8% (baseline 0–7%); 73–84% of it under real spots (random 7–23%).

REAL SAGE DATA | SYNTHETIC – UNVERIFIED | OUR RUN – MEASURED

results/demos/tomato/*/metrics.json, 3 leaves × 3 seeds, all 18 rig views. Colour classes fixed before the edits. The real underside is *unknown*: this shows the KB statement can be represented, not that it is right.

Ablations: Knowledge × Geometry

2×2 on total KB recall

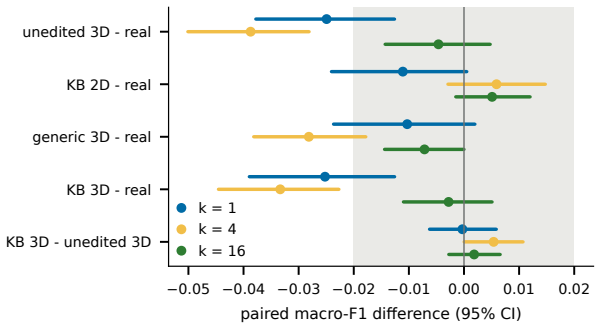
	no KB	KB
no geometry	D 0.00	K 0.17
geometry	G 0.04	H 0.53

Interaction $(H-G)-(K-D) = +0.32$, **n = 6** specimens with every cell (2 classes).

Ablation	Arms	Status
No KB / generic	B, E, G	✓
KB, wrong stage	C-NG	✓
Mismatched KB (neg. ctrl)	C-MM	✓
KB appearance only	K	✓
KB geometry only	J	✓
Appearance vs geometry 3D	K vs J vs H	✓
Placement: evidence vs random	J2/J2u, V/Vr, BLk/BLr	✓
Direction: cited vs opposite	CU/CD	✓
Viewpoints (1/2/4/8/adaptive)	–	✗
Severity, occlusion, background	–	✗

OUR RUN – MEASURED results/statistics/ablation_2x2.json. Total KB recall = mean of geometric and colour recall over the stage's required attributes.

E5: Minimal Downstream Test on Real Photos

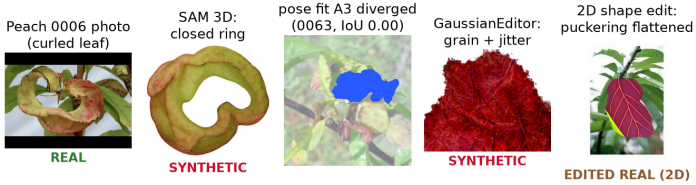


k real/class	KB-3D − real only	KB-3D − unedited 3D
1	-0.025	-0.000
4	-0.033	+0.005
16	-0.003	+0.002

Verdict (pre-registered): NOT SUPPORTED (falsified at this scale). The knowledge edit adds under 2 macro-F1 points over unedited renders; renders themselves hurt at small *k*. The render-to-photo gap comes before knowledge.

OUR RUN – MEASURED reports/14_downstream_minimal.md: within-crop disease classification, 5 crops, frozen DINOv2 + linear probe, 50 paired draws per *k*, 2,936 held-out real SAGE photos, pHash-disjoint. Macro-F1 differences; 95% CIs in results/downstream_minimal/contrasts.csv.

Failure Taxonomy



Reconstruction

Ring topology; 13/15 over-scaled; one-sheet leaves.

→ ✓ pose fix A3*; ✓ two-surface leaf (Test 3); ● reject closed rings before editing.

Optimisation

Pose refinement diverged on 3/15.

→ ✓ accept-if-better rule; ● add held-out depth as a second gate (worse on 6/15).

Editing

GE: grain, jitter, no shape; J blisters at random.

→ ✓ geometry editor for shape; ✓ evidence placement (weak gain); ● better lesion detector.

Biology

2D edit removes real puckering; stage mixing.

→ ✓ stage-aware KB; ● observed-morphology constraint; expert ratings blocked.

REAL SAGE DATA | SYNTHETIC – UNVERIFIED reports/09_failure_analysis.md; 00_full_failure_audit.md (root cause and status of each). ✓ done, ● next.

Execution Plan: Validate Before Generating More

01 Blinded plant-pathologist ratings
84-item package ready; 2–3 raters; one primary outcome declared first.

02 Small matched multi-view set
Corn smut and Tomato Leaf Mold first, with a ruler; held-out views score surface, direction, volume.

03 Independent geometry evaluation
Reference curl, gall volume and surface localisation, not the operators' own detectors.

04 Render-to-real diagnosis
Small matched pilot (background, lighting, texture, pose) before any 250 GPU-h E5.

05 Stronger baselines
KB-constrained 2D warp, two-layer 2D, shared-texture multi-view 2D.

Blocker	Unblocked by
No raters	3 recruited experts (01)
No matched real views	specimen capture (02)
No physical scale	ruler in captures
Render-to-photo gap	diagnosis (04)
Peach curl direction	a verbatim source

Question for you. Can we fund about 45 person-hours of capture and recruit three raters? Both precede any validity claim.

OUR RUN – MEASURED order follows the external review (validation first). Full blocker list and deliverables: backup slides; reports/12–15.

Go / No-Go After the Pilot

Decision	Required evidence vs what we have	Verdict
GO: KB-grounded 3D reconstruction	Verified hidden-trait recovery and a measured gain over KB 2D. *No held-out real views exist; minimal downstream test: no gain.*	✗ not met
PIVOT: 3D simulation and inspection	Plausible, controllable shape edits with limited claims. *Surface, direction and volume are representable and measured; consistent views; not yet expert-rated.*	● **best fit**
PIVOT: 2D knowledge-guided editing	Knowledge helps, 3D adds nothing. *Knowledge does steer 2D colour; but 2D cannot represent shape correctly (arm I).*	● partly
NO-GO: unreliable synthetic biology	Errors dominate. *Failures are real (ring, GE jitter) but diagnosed, reproducible and avoidable.*	✗ not dominant

Recommendation. Narrow the claim to *knowledge-controlled 3D symptom simulation*: an auditable KB→shape editor whose edits are traceable to cited text and consistent across views. Do not claim reconstruction of hidden anatomy. Numerical go/no-go thresholds will be set from pilot variances, not before.

OUR RUN – MEASURED evidence from slides E1–E4; decision logic from reports/10_final_experimental_protocol.md (Outcome D for now: no established advantage on real data).